

L03 Simulation Documentation

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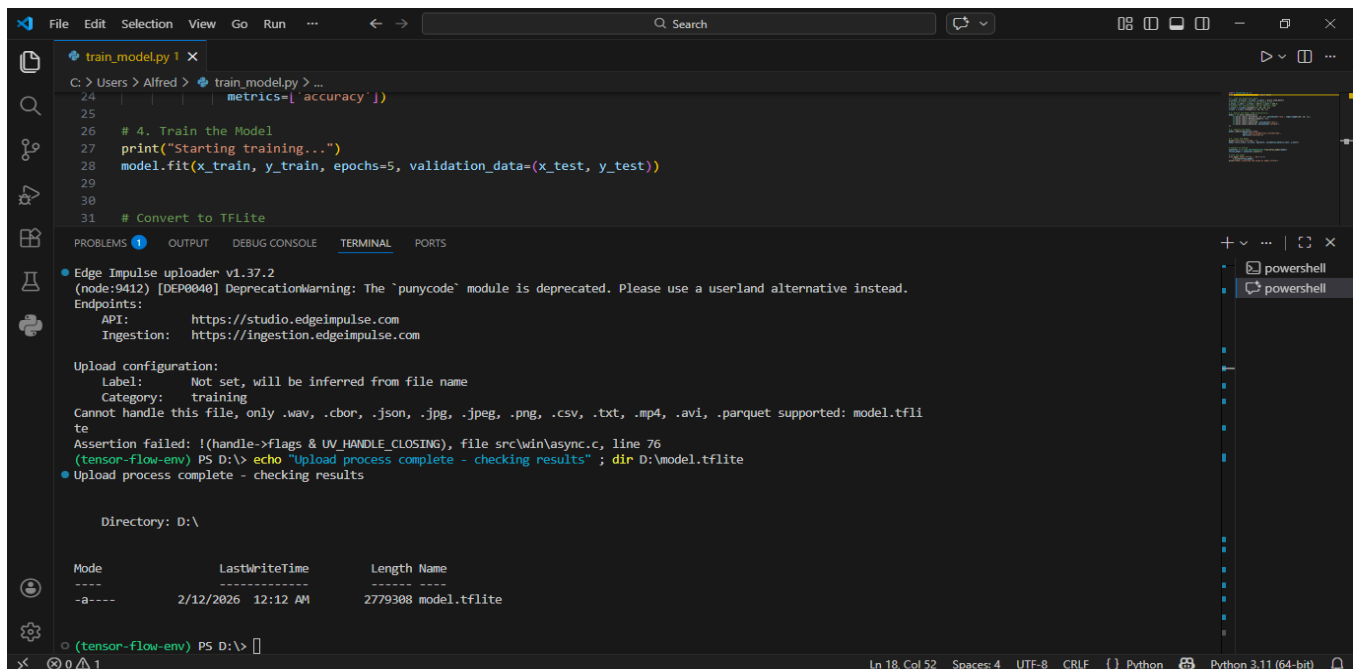
A.I. at the Edge & IIOT Env-17181

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1. Model Development (VS Code)

The project's first phase was about training a Convolutional Neural Network (CNN) on the MNIST digit dataset using Python. After completing the training, the model was turned into a TensorFlow Lite (.tflite) format for optimization for edge hardware. The process of conversion in the VS Code terminal confirmed a final model size of 2,779,308 bytes, which is about 2.7 MB.



The screenshot shows the VS Code interface with a Python file named `train_model.py` and a terminal window. The Python code includes a function `train_model` that trains a CNN on the MNIST dataset and converts it to TensorFlow Lite format. The terminal window shows the output of the training process, including the upload configuration and the final model size.

```

24 metrics=['accuracy'])
25
26 # 4. Train the Model
27 print("Starting training...")
28 model.fit(x_train, y_train, epochs=5, validation_data=(x_test, y_test))
29
30
31 # Convert to TFLite

```

Terminal Output:

```

Edge Impulse uploader v1.37.2
(node:9412) [DEP0040] DeprecationWarning: The `punycode` module is deprecated. Please use a userland alternative instead.
Endpoints:
  API:      https://studio.edgeimpulse.com
  Ingestion: https://ingestion.edgeimpulse.com

Upload configuration:
  Label:      Not set, will be inferred from file name
  Category:   training
Cannot handle this file, only .wav, .cbor, .json, .jpg, .jpeg, .png, .csv, .txt, .mp4, .avi, .parquet supported: model.tflite
Assertion failed: !(handle->flags & UV_HANDLE_CLOSING), file src\win\async.c, line 76
(tensor-flow-env) PS D:\> echo "Upload process complete - checking results" ; dir D:\model.tflite
Upload process complete - checking results

Directory: D:\

Mode                LastWriteTime         Length Name
----                -
-a-----         2/12/2026 12:12 AM       2779308 model.tflite

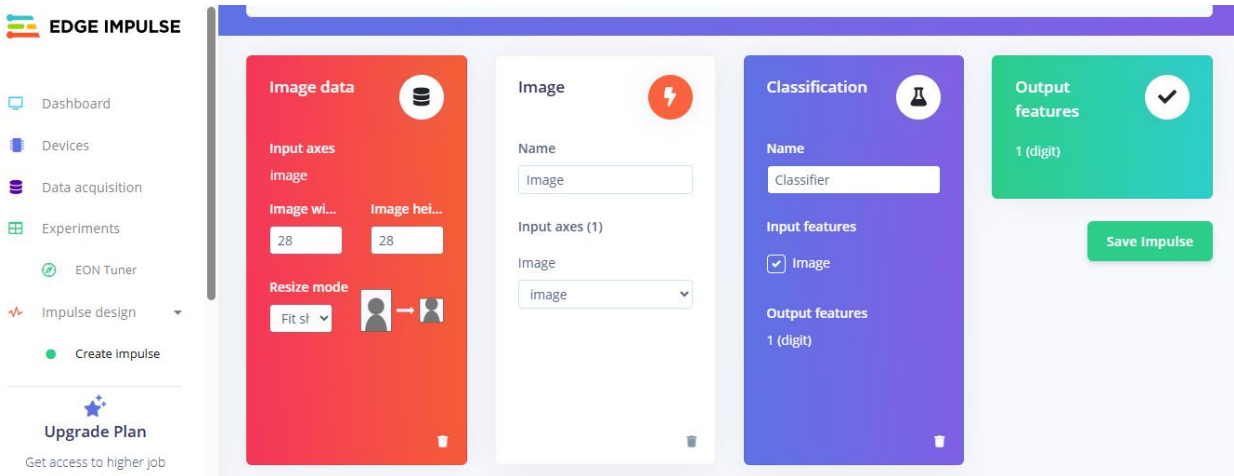
```

2. System Setup and Deployment Process

I uploaded the optimized model to the Edge Impulse platform for performance testing on a simulated microcontroller.

- ✓ **Configuration of the Input:** The impulse could handle 28x28 grayscale image data.
- ✓ **Impulse Blocks:** The design used an Image Processing block for the extraction of features as well as a Classification block for the identification of digits.

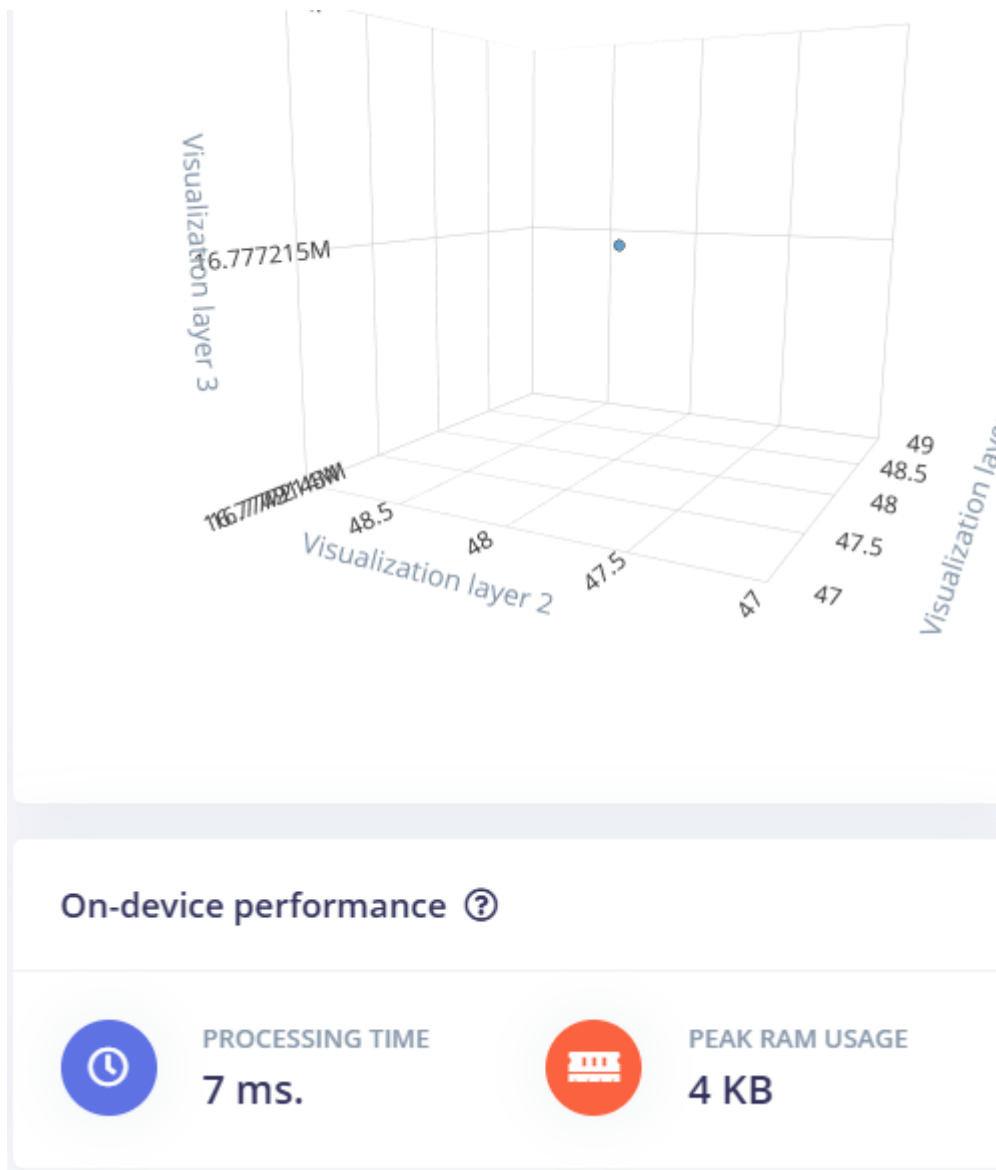
- ✓ **Data Preparation:** To facilitate simulation metrics, I uploaded sample images and labelled them manually as "digit".



3. Performance Validation Results

The hardware simulation was carried out with the key target being the Cortex-M4F 80MHz processor. The performance metrics shown below were captured during the phase of "Generate Features":

Metric	Simulated Result
Hardware Target	Cortex-M4F (80MHz)
Processing Time (Latency)	7 ms
Peak RAM Usage	4 KB
Flash Usage (Model Size)	2.7 MB



4. Conclusion

The simulation was successful in validating that the model of MNIST is appropriate for deployment on devices of low-power edge. In just 7 ms, the model was capable of performing real-time digit recognition within the Cortex-M4F architecture's memory constraints.

